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Memory circuitry and method used in forming memory circuitry

Abstract

A method used in forming memory circuitry comprises forming a stack comprising vertically-alternating first tiers and second tiers. The stack extends from a memory-array region into a stair-step region. The stair-step region comprises a cavity comprising a flight of stairs. The first tiers are conductive and the second tiers are insulative at least in a finished-circuitry construction. A lining is formed in and that less-than-fills the cavity atop treads of the stairs. Individual of the treads comprise conducting material of one of the first tiers in the finished-circuitry construction. The lining that is atop the treads is replaced with at least one of metal material, polysilicon, or SiGe and insulative material is provided in remaining volume of the cavity directly above the at least one of the metal material, the polysilicon, or the SiGe. Conductive vias are formed through the insulative material and the at least one of the metal material, the polysilicon, or the SiGe. Individual of the conductive vias are directly above and directly against the conducting material of one of the individual treads. Other embodiments, including structure, are disclosed.

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Background/Summary

TECHNICAL FIELD

(1) Embodiments disclosed herein pertain to memory circuitry and to methods used in forming memory circuitry.

BACKGROUND

(2) Memory is one type of integrated circuitry and is used in computer systems for storing data. Memory may be fabricated in one or more arrays of individual memory cells. Memory cells may be written to, or read from, using digitlines (which may also be referred to as bitlines, data lines, or sense lines) and access lines (which may also be referred to as wordlines). The sense lines may conductively interconnect memory cells along columns of the array, and the access lines may conductively interconnect memory cells along rows of the array. Each memory cell may be uniquely addressed through the combination of a sense line and an access line.

(3) Memory cells may be volatile, semi-volatile, or non-volatile. Non-volatile memory cells can store data for extended periods of time in the absence of power. Non-volatile memory is conventionally specified to be memory having a retention time of at least about 10 years. Volatile memory dissipates and is therefore refreshed/rewritten to maintain data storage. Volatile memory may have a retention time of milliseconds or less. Regardless, memory cells are configured to retain or store memory in at least two different selectable states. In a binary system, the states are considered as either a “0” or a “1”. In other systems, at least some individual memory cells may be configured to store more than two levels or states of information.

(4) A field effect transistor is one type of electronic component that may be used in a memory cell. These transistors comprise a pair of conductive source/drain regions having a semiconductive channel region there-between. A conductive gate is adjacent the channel region and separated there-from by a thin gate insulator. Application of a suitable voltage to the gate allows current to flow from one of the source/drain regions to the other through the channel region. When the voltage is removed from the gate, current is largely prevented from flowing through the channel region. Field effect transistors may also include additional structure, for example a reversibly programmable charge-storage region as part of the gate construction between the gate insulator and the conductive gate.
(5) Flash memory is one type of memory and has numerous uses in modern computers and devices. For instance, modern personal computers may have BIOS stored on a flash memory chip. As another example, it is becoming increasingly common for computers and other devices to utilize flash memory in solid state drives to replace conventional hard drives. As yet another example, flash memory is popular in wireless electronic devices because it enables manufacturers to support new communication protocols as they become standardized, and to provide the ability to remotely upgrade the devices for enhanced features.
(6) NAND may be a basic architecture of integrated flash memory. A NAND cell unit comprises at least one selecting device coupled in series to a serial combination of memory cells (with the serial combination commonly being referred to as a NAND string). NAND architecture may be configured in a three-dimensional arrangement comprising vertically-stacked memory cells individually comprising a reversibly programmable vertical transistor. Control or other circuitry may be formed below the vertically-stacked memory cells. Other volatile or non-volatile memory array architectures may also comprise vertically-stacked memory cells that individually comprise a transistor.
(7) Memory arrays may be arranged in memory pages, memory blocks and partial blocks (e.g., sub-blocks), and memory planes, for example as shown and described in any of U.S. Patent Application Publication Nos. 2015/0228651, 2016/0267984, and 2017/0140833. The memory blocks may at least in part define longitudinal outlines of individual wordlines in individual wordline tiers of vertically-stacked memory cells. Connections to these wordlines may occur in a so-called “stair-step structure” at an end or edge of an array of the vertically-stacked memory cells. The stair-step structure includes individual “stairs” (alternately termed “steps” or “stair-steps”) that define contact regions of the individual wordlines upon which elevationally-extending conductive vias contact to provide electrical access to the wordlines.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagrammatic view of a portion of memory circuitry in process in accordance with embodiments of the invention.
(2) FIG. **3** is a diagrammatic cross-sectional view taken through line **3-3** in FIG. **1**.
(3) FIGS. **2** and **4-47** are diagrammatic sectional, expanded, enlarged, and/or partial views of the construction of FIGS. **1-3** or portions thereof, and/or of alternate embodiments.
DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS
(4) Embodiments of the invention encompass methods used in forming memory circuitry comprising a memory array, for example an array of NAND or other memory cells that may have at least some peripheral control circuitry under the array (e.g., CMOS-under-array). Embodiments of the invention encompass so-called “gate-last” or “replacement-gate” processing, so-called “gate-first” processing, and other processing whether existing or future-developed independent of when transistor gates are formed. Embodiments of the invention also encompass integrated circuitry

comprising a memory array comprising strings of memory cells (e.g., NAND architecture) independent of method of manufacture. Some example embodiments are described with reference to FIGS. **1-47**.

(5) FIGS. **1-8** show an example construction **10** having two memory-array regions **12** in which elevationally-extending strings of transistors and/or memory cells will be formed. The two memory-array regions **12** may be of the same or different constructions relative one another. In one embodiment, a stair-step region **13** is between memory-array regions **12** and comprises stair-step structures as described below. Alternately, by way of example, a stair-step region may be at the end of a single memory-array region (not shown). FIGS. **6-8** are of different and varying scales compared to FIGS. **1-5** for clarity in disclosure more pertinent to stair-step region **13** than to memory-array regions **12**. Example construction **10** comprises a base substrate **11** having any one or more of conductive/conductor/conducting, semiconductive/semiconductor/semiconducting, or insulative/insulator/insulating (i.e., electrically herein) materials. Various materials have been formed elevationally over base substrate **11**. Materials may be aside, elevationally inward, or elevationally outward of the FIGS. **1-8**—depicted materials. For example, other partially or wholly fabricated components of integrated circuitry may be provided somewhere above, about, or within base substrate **11**. Control and/or other peripheral circuitry for operating components within an array (e.g., individual array regions **12**) of elevationally-extending strings of memory cells may also be fabricated and may or may not be wholly or partially within an array or sub-array. Further, multiple sub-arrays may also be fabricated and operated independently, in tandem, or otherwise relative one another. In this document, a “sub-array” may also be considered as an array.

(6) A conductor tier **16** comprising conductor material **17** (e.g., WSi.sub.x under conductively-doped polysilicon) is above substrate **11**. Conductor tier **16** may comprise part of control circuitry (e.g., peripheral-under-array circuitry and/or a common source line or plate) used to control read and write access to the transistors and/or memory cells in array **12**. A vertical stack **18** comprising vertically-alternating insulative tiers **20** and conductive tiers **22** is directly above conductor tier **16** and extends from memory-array region **12** into stair-step region **13**. In some embodiments, conductive tiers **22** may be referred to as first tiers **22** and insulative tiers **20** may be referred to as second tiers **20**, with first tiers **22** being conductive and second tiers **20** being insulative at least in a finished-circuitry construction. Example thickness for each of tiers **20** and **22** is 20 to 60 nanometers. The example uppermost tier **20** may be thicker/thickest compared to one or more other tiers **20** and/or **22**. Example first tiers **22** comprise material **26** (e.g., silicon nitride) and example second tiers **20** comprise material **24** (e.g., silicon dioxide). Only a small number of tiers **20** and **22** is shown in FIGS. **2-5** and other figures, with more likely stack **18** comprising dozens, a hundred or more, etc. of tiers **20** and **22**. Other circuitry that may or may not be part of peripheral and/or control circuitry may be between conductor tier **16** and stack **18**. For example, multiple vertically-alternating tiers of conductive material and insulative material of such circuitry may be below a lowest of the conductive tiers **22** and/or above an uppermost of the conductive tiers **22**. For example, one or more select gate tiers (not shown) may be between conductor tier **16** and the lowest conductive tier **22** and one or more select gate tiers may be above an uppermost of conductive tiers **22** (not shown). Alternately or additionally, at least one of the depicted uppermost and lowest conductive tiers **22** may be a select gate tier.

(7) Channel openings **25** have been formed (e.g., by etching) through insulative tiers **20** and conductive tiers **22** to conductor tier **16**. Channel openings **25** may taper radially-inward and/or radially-outward (not shown) moving deeper in stack **18**. In some embodiments, channel openings **25** may go into conductor material **17** of conductor tier **16** as shown or may stop there-atop (not shown). Alternately, as an example, channel openings **25** may stop atop or within the lowest insulative tier **20**. A reason for extending channel openings **25** at least to conductor material **17** of conductor tier **16** is to assure direct electrical coupling of channel material to conductor tier **16** without using alternative processing and structure to do so when such a connection is desired

and/or to provide an anchoring effect to material that is within channel openings **25**. Etch-stop material (not shown) may be within or atop conductor material **17** of conductor tier **16** to facilitate stopping of the etching of channel openings **25** relative to conductor tier **16** when such is desired. Such etch-stop material may be sacrificial or non-sacrificial. By way of example and for brevity only, channel openings **25** are shown as being arranged in groups or columns of staggered rows of four and five openings **25** per row and being arrayed in laterally-spaced memory-block regions **58** that will comprise laterally-spaced memory blocks **58** in a finished circuitry construction. In this document, “block” is generic to include “sub-block”. Memory-block regions **58** and resultant memory blocks **58** (not yet shown) may be considered as being longitudinally elongated and oriented, for example along a first direction **55**, with a second direction **99** being orthogonal thereto. Any alternate existing or future-developed arrangement and construction may be used.

(8) Transistor channel material may be formed in the individual channel openings elevationally along the insulative tiers and the conductive tiers, thus comprising individual channel-material strings, which is directly electrically coupled with conductive material in the conductor tier. Individual memory cells of the example memory array being formed may comprise a gate region (e.g., a control-gate region) and a memory structure laterally between the gate region and the channel material. In one such embodiment, the memory structure is formed to comprise a charge-blocking region, storage material (e.g., charge-storage material), and an insulative charge-passage material. The storage material (e.g., floating gate material such as doped or undoped silicon or charge-trapping material such as silicon nitride, metal dots, etc.) of the individual memory cells is elevationally along individual of the charge-blocking regions. The insulative charge-passage material (e.g., a band gap-engineered structure having nitrogen-containing material [e.g., silicon nitride] sandwiched between two insulator oxides [e.g., silicon dioxide]) is laterally between the channel material and the storage material.

(9) The figures show one embodiment wherein charge-blocking material **30**, storage material **32**, and charge-passage material **34** have been formed in individual channel openings **25** elevationally along insulative tiers **20** and conductive tiers **22**. Transistor materials **30**, **32**, and **34** (e.g., memory-cell materials) may be formed by, for example, deposition of respective thin layers thereof over stack **18** and within individual channel openings **25** followed by planarizing such back at least to a top surface of stack **18** as shown.

(10) Channel material **36** has also been formed in channel openings **25** elevationally along insulative tiers **20** and conductive tiers **22** and comprise individual channel-material strings **53** in one embodiment having memory-cell materials (e.g., **30**, **32**, and **34**) there-along and with material **24** in insulative tiers **20** being horizontally-between immediately-adjacent channel-material strings **53**. Materials **30**, **32**, **34**, and **36** are collectively shown as and only designated as material **37** in some figures due to scale. Example channel materials **36** include appropriately-doped crystalline semiconductor material, such as one or more silicon, germanium, and so-called III/V semiconductor materials (e.g., GaAs, InP, GaP, and GaN). Example thickness for each of materials **30**, **32**, **34**, and **36** is 25 to 100 Angstroms. Punch etching may be conducted as shown to remove materials **30**, **32**, and **34** from the bases of channel openings **25** to expose conductor tier **16** such that channel material **36** (channel-material string **53**) is directly electrically coupled with conductor material **17** of conductor tier **16**. Such punch etching may occur separately with respect to each of materials **30**, **32**, and **34** (as shown) or may occur collectively with respect to all after deposition of material **34** (not shown). Alternately, and by way of example only, no punch etching may be conducted and channel material **36** may be directly electrically coupled with conductor material **17** of conductor tier **16** by a separate conductive interconnect (not shown). Channel openings **25** are shown as comprising a radially-central solid dielectric material **38** (e.g., spin-on-dielectric, silicon dioxide, and/or silicon nitride). Alternately, and by way of example only, the radially-central portion within channel openings **25** may include void space(s) (not shown) and/or be devoid of solid material (not shown).

(11) Referring to FIGS. **1** and **6-8**, and in one embodiment, an array of cavities **66** has been formed in stair-step region **13** and that individually comprise a stair-step structure as described below. Example cavities **66** are aligned longitudinally end-to-end in individual memory-block regions **58** and have a crest **81** between immediately-adjacent cavities **66**. Alternately, only a single cavity may be in individual memory-block regions **58** (not shown). Nevertheless, method and structure embodiments include fabrication of and a resultant construction having only a single cavity **66** and the discussion largely proceeds with respect to a single cavity **66**. Cavities **66** are shown as being rectangular in horizontal cross-section, although other shape(s) may be used and all need not be of the same shape relative one another. For brevity, less tiers **20** and **22** are shown in FIGS. **3** and **5** as compared to FIGS. **7** and **8**, with more tiers **20** and **22** being shown in FIGS. **7** and **8** for clarity and for better emphasis of an example depth of cavities **66** and processing/aspects associated therewith.
(12) Cavities **66** individually comprise a flight **67** or **69** of stairs **70** in a first vertical cross-section (e.g., that of FIG. **7**) along a first direction (e.g., **55**). Flights **67** and **69** with a landing **91** there-between comprise a stair-step structure. Example flights **67** and **69** oppose one another in cavity **66** and individual stairs **70** comprise a tread **75** and a riser **85**. Individual treads **75** will comprise conducting material of one of conductive tiers **22** in the finished-circuitry construction. Cavity **66** with flights **67** and **69** may be formed by any existing or later-developed method(s). As one such example, a masking material (e.g., a photo-imageable material such as photoresist) may be formed atop stack **18** and an opening formed there-through. Then, the masking material may be used as a mask while etching (e.g., anisotropically) through the opening to extend such opening into at least two outermost two tiers **20**, **22**. The resultant construction may then be subjected to a successive alternating series of lateral-trimming etches of the masking material followed by etching deeper into stack **18**, two-tiers **20**, **22** at a time, using the trimmed masking material having a successively widened opening as a mask. Such an example may result in the forming of flight **67** into stack **18** that comprises vertically alternating tiers **20**, **22** of different composition materials **24**, **26**, and in the forming of another flight **69** opposite and facing flight **67** (e.g., in mirror-image and as shown). Likely more stairs **70** will be in flights **67** and/or **69** than shown. Example stairs **70** in stack **18** are individually shown as comprising one first tier **22** and one second tier **20**. More first and second tiers per stair **70** may be used, for example if forming multiple treads per stair (e.g., along second direction **99** and not shown).
(13) In one example, one of two opposing flights **67** and **69** is operative (e.g., flight **67**) and the other of two opposing flights **67** and **69** is dummy (e.g., flight **69**) in the finished-circuitry construction. In this document, a flight that is "dummy" is circuit-inoperative having stairs thereof in which no current flows in conductive material of the steps and which may be a circuit-inoperable dead end that is not part of a current flow path of a circuit even if extending to or from an electronic component. When inoperative, position of operative vs. inoperative relative to flights **67** and **69** may of course be reversed. Multiple operative flights and multiple dummy flights may be formed in multiple cavities **66**, for example longitudinally end-to-end as shown and to different depths within stack **18**. Pairs of opposing mirror-image operative and dummy flights may be considered as defining a stadium (e.g., a vertically recessed portion having opposing flights of stairs as shown). Alternately, only a single flight **67** or **69** may be formed (not shown) in one or more individual cavities **66**. Regardless, cavities **66** may be formed before or after forming channel-material strings **53**. Cavities **66** may be considered as having sidewalls **71**, **88**, and **85** (risers **85** along with sidewalls **88** effectively being part of the sidewalls of cavities **66** that are along second direction **99**), with sidewalls **71** being along first direction **55**. Sidewalls **71**, **88**, and/or **85** may taper laterally-inward moving deeper into stack **18** (not shown).
(14) Referring to FIGS. **9-11**, a lining **64** comprising material **59** has been formed in and less-than-fills cavity **66** atop treads **75** of stairs **70**. In one embodiment, lining **64** (material **59**) comprises silicon nitride.
(15) Referring to FIGS. **12-14**, insulative material **76** (e.g., silicon dioxide) has been formed in

remaining volume of cavity **66** directly above lining **64**.

(16) Referring to FIGS. **15-21**, horizontally-elongated trenches **40** have been formed into stack **18** (e.g., by anisotropic etching) and which are individually between immediately-laterally-adjacent memory-block regions **58**. Trenches **40** will typically be wider than channel openings **25** (e.g., 3 to 10 times wider). Trenches **40** may have respective bottoms that are directly against conductor material **17** (e.g., atop or within) of conductor tier **16** (as shown) or may have respective bottoms that are above conductor material **17** of conductor tier **16** (not shown). Trenches **40** may taper laterally-inward and/or outward in vertical cross-section (not shown). Sidewalls **71** of cavities **66** may be laterally-spaced inwardly from immediately-laterally-adjacent trenches **40** or may not be so spaced, for example depending on whether operative stair flight **67** is directly electrically coupled to only one or to both of two memory-array regions **12**. In one embodiment, through-array-via (TAV) openings **68** have also been formed through insulative material **76** and treads **75**, with there being multiple TAV openings **68** per and extending through individual treads **75** (six being shown) in stair-flight **67**. TAV openings **68** may be formed before, after, or commensurate with forming of trenches **40**. Other TAV openings may be formed (not shown), for example through one or more of landing **91**, stairs **70** of flight **69**, and crest(s) **81**.

(17) Referring to FIGS. **22-24**, and in one embodiment, through TAV openings **68**, silicon nitride **59** of lining **64** and silicon nitride **26** of first tiers **22** have been isotropically etched laterally-outward relative to TAV openings **68**. In the example embodiment, silicon nitride **59** of lining **64** has been so laterally etched to a greater degree than silicon nitride **26** of first tiers **22** and wherein such etching horizontally interconnects multiple TAV openings **68** into a single opening **72** per tread **75** where etched silicon nitride **59** of lining **64** was. The artisan is capable of using different deposition techniques for forming silicon nitride **59** to etch faster than silicon nitride **26** in a given etching chemistry, for example an etching chemistry at least predominantly comprising phosphoric acid (e.g., by depositing silicon nitride **59** at a much lower deposition temperature than that at which silicon nitride **26** is deposited). The example etching optionally shows etching of silicon nitride **59** to riser **85** of individual stairs **70** although less degree of isotropic etching may occur over and/or as part of individual treads **75**. Trenches **40** may be masked (not shown) during such etching.

(18) Referring to FIGS. **25-27**, through the TAV openings **68**, at least one of metal material **73**, polysilicon **73**, or SiGe **73** (including any mixture of two or more of such) has been formed in single opening **72** where etched silicon nitride **59** of lining **64** was and across horizontal locations **74** where conductive vias to conducting material (neither being shown yet) of individual treads **75** will be formed. That which is **73** (the at least one) is also shown as being formed vertically all along TAV openings **68** and in laterally-expanded portions of TAV openings **68** in first tiers **22** (when such are so formed). Regardless, in one embodiment the at least one **73** comprises metal material, in one embodiment comprises polysilicon, and in one embodiment comprises SiGe. In one embodiment, the at least one **73** is conductive (e.g., an elemental metal [W], an alloy or two or more elemental metals, and/or a conductive metallic compound [TiN] as example conductive metal materials, conductively-doped polysilicon, and/or conductively-doped SiGe) and in another embodiment is insulative (e.g., Al.sub.2O.sub.3 and HfO.sub.2 as example insulative metal materials).

(19) Referring to FIGS. **28** and **29** and in one embodiment, material **73** (i.e., the at least one of the metal material, the polysilicon, or the SiGe) has been wet isotropically etched to remove it from being vertically-continuous within TAV openings **68** (when initially so-continuous) and to laterally-recess material **73** back relative to sidewalls of TAV openings **68**. Thereafter, and regardless, an insulative lining **77** (e.g., silicon dioxide and/or silicon nitride) and a conductive core **78** (e.g., W radially-inside of a TiN lining) of individual TAV constructions **79** have been formed. In one embodiment and as shown, there are multiple TAV constructions **79** per tread **75** and, in one such embodiment as shown, material **73** is conductive and is horizontally-continuous among the

multiple TAV constructions **79** in the respective tread **75**.

(20) Referring to FIGS. **30-39**, material **26** (not shown) of first tiers **22** has been removed, for example by being isotropically etched away through trenches **40** ideally selectively relative to the other exposed materials (e.g., using liquid or vapor H.sub.3PO.sub.4 as a primary etchant where material **26** is silicon nitride and other materials comprise one or more oxides or polysilicon). Material **26** (not shown) in conductive tiers **22** in the example embodiment is sacrificial and has been replaced with conducting material **48**, and which has thereafter been removed from trenches **40**, thus forming individual conductive lines **29** (e.g., wordlines in stack **18**) and elevationally-extending strings **49** of individual transistors and/or memory cells **56** in stack **18**.

(21) A thin insulative liner (e.g., Al.sub.2O.sub.3 and not shown) may be formed before forming conducting material **48**. Approximate locations of transistors and/or memory cells **56** are indicated with a bracket in some figures and some with dashed outlines in some figures, with transistors and/or memory cells **56** being essentially ring-like or annular in the depicted example. Alternately, transistors and/or memory cells **56** may not be completely encircling relative to individual channel openings **25** such that each channel opening **25** may have two or more elevationally-extending strings **49** (e.g., multiple transistors and/or memory cells about individual channel openings in individual conductive tiers with perhaps multiple wordlines per channel opening in individual conductive tiers, and not shown). Conducting material **48** may be considered as having terminal ends **50** corresponding to control-gate regions **52** of individual transistors and/or memory cells **56**. Control-gate regions **52** in the depicted embodiment comprise individual portions of individual conductive lines **29**. Materials **30**, **32**, and **34** may be considered as a memory structure **65** that is laterally between control-gate region **52** and channel material **36**. In one embodiment and as shown with respect to the example “gate-last” processing, conducting material **48** of conductive tiers **22** is formed after forming channel openings **25** and/or trenches **40**. Alternately, the conducting material of the conductive tiers may be formed before forming channel openings **25** and/or trenches **40** (not shown), for example with respect to “gate-first” processing.

(22) A charge-blocking region (e.g., charge-blocking material **30**) is between storage material **32** and individual control-gate regions **52**. A charge block may have the following functions in a memory cell: In a program mode, the charge block may prevent charge carriers from passing out of the storage material (e.g., floating-gate material, charge-trapping material, etc.) toward the control gate, and in an erase mode the charge block may prevent charge carriers from flowing into the storage material from the control gate. Accordingly, a charge block may function to block charge migration between the control-gate region and the storage material of individual memory cells. An example charge-blocking region as shown comprises insulator material **30**. By way of further examples, a charge-blocking region may comprise a laterally (e.g., radially) outer portion of the storage material (e.g., material **32**) where such storage material is insulative (e.g., in the absence of any different-composition material between an insulative storage material **32** and conducting material **48**). Regardless, as an additional example, an interface of a storage material and conductive material of a control gate may be sufficient to function as a charge-blocking region in the absence of any separate-composition-insulator material **30**. Further, an interface of conducting material **48** with material **30** (when present) in combination with insulator material **30** may together function as a charge-blocking region, and as alternately or additionally may a laterally-outer region of an insulative storage material (e.g., a silicon nitride material **32**). An example material **30** is one or more of silicon hafnium oxide and silicon dioxide.

(23) Intervening material **57** has been formed in trenches **40** and thereby laterally-between and longitudinally-along immediately-laterally-adjacent memory blocks **58**. Intervening material **57** may provide lateral electrical isolation (insulation) between immediately-laterally-adjacent memory blocks. Such may include one or more of insulative, semiconductive, and conducting materials and, regardless, may facilitate conductive tiers **22** from shorting relative one another in a finished circuitry construction. Example insulative materials are one or more of SiO.sub.2, Si.sub.3N.sub.4,

and Al.sub.2O.sub.3. Intervening material **57** may include through-array vias (not shown).
(24) Referring to FIGS. **40** and **41** and in one embodiment, first etching has been conducted through insulative material **76** to form conductive via openings **82** in horizontal locations **74** using material **73** as an etch-stop during the first etching such that conductive via openings **82** stop atop or within material **73**. Reference to "first etching" is in the context of being before a "second etching" referred to below and is not necessarily the first-in-time etching of insulative material **76** or openings **82** therein and additional etching may occur between the stated first etching and the second etching.
(25) Referring to FIG. **42** and after the first etching, second etching has been conducted through material **73** to extend conductive via openings **82** deeper. Such second etching may be wet and isotropic (not shown) or may be dry and anisotropic (as shown). FIG. **43** shows subsequent processing (e.g., anisotropic etching of material **24**) whereby conducting material **48** of individual treads **75** has been exposed.
(26) FIGS. **44-47** show subsequent processing whereby conductive vias **80** have been formed in extended conductive via openings **82**. Individual conductive vias **80** are directly above and directly against conducting material **48** of individual treads **75**. TAV constructions **79** may be directly electrically coupled to conductive vias **80** and/or other circuitry, as may be conductive vias **80**, that is not particularly pertinent to aspects of the inventions disclosed herein. An insulative lining (e.g., silicon nitride and/or silicon dioxide) may be about conductive vias **80** as shown (not numerically designated).
(27) In one embodiment, material **73** and conducting material **48** of the one first tier **22** that is directly below material **73** are of different compositions relative one another. In embodiment, material **73** is conductive and conducting material **48** of the one first tier **22** that is directly below material **73** is of different composition from that of material **73** and in another such embodiment is of the same composition as that of material **73**.
(28) In one embodiment and as shown, individual treads **75** in the finished-circuitry construction comprise material **73**, conducting material **48** of one of first tiers **22** directly below material **73**, and an intervening insulating material (e.g., **24**) vertically-between material **73** and conducting material **48** of the one first tier **22** that is directly below material **73**. In one such embodiment, TAV constructions **79** have been formed to extend through insulative material **76**, through material **73**, through intervening insulating material **24**, and conducting material **48** of the one first tier **22** that is directly below material **73** of individual treads **75**.
(29) Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used in the embodiments shown and described with reference to the above embodiments.
(30) In one embodiment, a method used in forming memory circuitry (e.g., **10**) comprises forming a stack (e.g., **18**) comprising vertically-alternating first tiers (e.g., **22**) and second tiers (e.g., **20**). The stack extends from a memory-array region (e.g., **12**) into a stair-step region (e.g., **13**). The stair-step region comprises a cavity (e.g., **66**) comprising a flight (e.g., **67**) of stairs (e.g., **70**). The first tiers are conductive and the second tiers are insulative at least in a finished-circuitry construction. A lining (e.g., **64**) is formed in and less-than-fills the cavity atop treads (e.g., **75**) of the stairs. Individual of the treads comprise conducting material (e.g., **48**) of one of the first tiers in the finished-circuitry construction. The lining that is atop the treads is replaced with at least one of metal material (e.g., **73**), polysilicon (e.g., **73**), or SiGe (e.g., **73**) and insulative material (e.g., **76**) is provided in remaining volume of the cavity directly above the at least one of the metal material, the polysilicon, or the SiGe. Conductive vias (e.g., **80**) are formed through the insulative material and the at least one of the metal material, the polysilicon, or the SiGe. Individual of the conductive vias are directly above and directly against the conducting material of one of the individual treads. In one such embodiment, the providing of the insulative material in the remaining volume of the cavity occurs before the replacing of the lining with the at least one of the metal material, the

polysilicon, or the SiGe. Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

(31) Forming and using a material **73** as described herein may provide better etch-stopping function when forming conductive via openings **82** compared to prior methods.

(32) Alternate embodiment constructions may result from method embodiments described above, or otherwise. Regardless, embodiments of the invention encompass memory arrays independent of method of manufacture. Nevertheless, such memory arrays may have any of the attributes as described herein in method embodiments. Likewise, the above-described method embodiments may incorporate, form, and/or have any of the attributes described with respect to device embodiments.

(33) In one embodiment, memory circuitry (e.g., **10**) comprising strings (e.g., **49**) of memory cells (e.g., **56**) comprises a stack (e.g., **18**) comprising vertically-alternating insulative tiers (e.g., **20**) and conductive tiers (e.g., **22**). Channel-material strings (e.g., **53**) of memory cells (e.g., **56**) extend through the insulative tiers and the conductive tiers in a memory-array region (e.g., **12**). The insulative tiers and the conductive tiers extend from the memory-array region into a stair-step region (e.g., **13**). The stair-step region comprises a cavity (e.g., **66**) comprising a flight (e.g., **67**) of stairs (e.g., **70**). Treads (e.g., **75**) of individual of the stairs comprise conducting material (e.g., **48**) of one of the conductive tiers. Individual of the treads comprise: at least one of metal material (e.g., **73**), polysilicon (e.g., **73**), or SiGe (e.g., **73**); the conducting material of one of the conductive tiers directly below the at least one of the metal material, the polysilicon, or the SiGe; and an intervening insulating material (e.g., **24**) vertically-between the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one conductive tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe.

(34) Insulative material (e.g., **76**) is in the cavity directly above the treads. Conductive vias (e.g., **80**) extend through the insulative material. Individual of the conductive vias extend through the at least one of the metal material, the polysilicon, or the SiGe and the intervening insulating material to be directly against the conducting material of one of the individual treads. Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

(35) The above processing(s) or construction(s) may be considered as being relative to an array of components formed as or within a single stack or single deck of such components above or as part of an underlying base substrate (albeit, the single stack/deck may have multiple tiers). Control and/or other peripheral circuitry for operating or accessing such components within an array may also be formed anywhere as part of the finished construction, and in some embodiments may be under the array (e.g., CMOS under-array). Regardless, one or more additional such stack(s)/deck(s) may be provided or fabricated above and/or below that shown in the figures or described above. Further, the array(s) of components may be the same or different relative one another in different stacks/decks and different stacks/decks may be of the same thickness or of different thicknesses relative one another. Intervening structure may be provided between immediately-vertically-adjacent stacks/decks (e.g., additional circuitry and/or dielectric layers). Also, different stacks/decks may be electrically coupled relative one another. The multiple stacks/decks may be fabricated separately and sequentially (e.g., one atop another), or two or more stacks/decks may be fabricated at essentially the same time.

(36) The assemblies and structures discussed above may be used in integrated circuits/circuitry and may be incorporated into electronic systems. Such electronic systems may be used in, for example, memory modules, device drivers, power modules, communication modems, processor modules, and application-specific modules, and may include multilayer, multichip modules. The electronic systems may be any of a broad range of systems, such as, for example, cameras, wireless devices, displays, chip sets, set top boxes, games, lighting, vehicles, clocks, televisions, cell phones, personal computers, automobiles, industrial control systems, aircraft, etc.

(37) In this document unless otherwise indicated, “elevational”, “higher”, “upper”, “lower”, “top”,

"atop", "bottom", "above", "below", "under", "beneath", "up", and "down" are generally with reference to the vertical direction. "Horizontal" refers to a general direction (i.e., within 10 degrees) along a primary substrate surface and may be relative to which the substrate is processed during fabrication, and vertical is a direction generally orthogonal thereto. Reference to "exactly horizontal" is the direction along the primary substrate surface (i.e., no degrees there-from) and may be relative to which the substrate is processed during fabrication. Further, "vertical" and "horizontal" as used herein are generally perpendicular directions relative one another and independent of orientation of the substrate in three-dimensional space. Additionally, "elevationally-extending" and "extend(ing) elevationally" refer to a direction that is angled away by at least 45° from exactly horizontal. Further, "extend(ing) elevationally", "elevationally-extending", "extend(ing) horizontally", "horizontally-extending" and the like with respect to a field effect transistor are with reference to orientation of the transistor's channel length along which current flows in operation between the source/drain regions. For bipolar junction transistors, "extend(ing) elevationally" "elevationally-extending", "extend(ing) horizontally", "horizontally-extending" and the like, are with reference to orientation of the base length along which current flows in operation between the emitter and collector. In some embodiments, any component, feature, and/or region that extends elevationally extends vertically or within 10° of vertical.

(38) Further, "directly above", "directly below", and "directly under" require at least some lateral overlap (i.e., horizontally) of two stated regions/materials/components relative one another. Also, use of "above" not preceded by "directly" only requires that some portion of the stated region/material/component that is above the other be elevationally outward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components). Analogously, use of "below" and "under" not preceded by "directly" only requires that some portion of the stated region/material/component that is below/under the other be elevationally inward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components).

(39) Any of the materials, regions, and structures described herein may be homogenous or non-homogenous, and regardless may be continuous or discontinuous over any material which such overlie. Where one or more example composition(s) is/are provided for any material, that material may comprise, consist essentially of, or consist of such one or more composition(s). Further, unless otherwise stated, each material may be formed using any suitable existing or future-developed technique, with atomic layer deposition, chemical vapor deposition, physical vapor deposition, epitaxial growth, diffusion doping, and ion implanting being examples.

(40) Additionally, "thickness" by itself (no preceding directional adjective) is defined as the mean straight-line distance through a given material or region perpendicularly from a closest surface of an immediately-adjacent material of different composition or of an immediately-adjacent region. Additionally, the various materials or regions described herein may be of substantially constant thickness or of variable thicknesses. If of variable thickness, thickness refers to average thickness unless otherwise indicated, and such material or region will have some minimum thickness and some maximum thickness due to the thickness being variable. As used herein, "different composition" only requires those portions of two stated materials or regions that may be directly against one another to be chemically and/or physically different, for example if such materials or regions are not homogenous. If the two stated materials or regions are not directly against one another, "different composition" only requires that those portions of the two stated materials or regions that are closest to one another be chemically and/or physically different if such materials or regions are not homogenous. In this document, a material, region, or structure is "directly against" another when there is at least some physical touching contact of the stated materials, regions, or structures relative one another. In contrast, "over", "on", "adjacent", "along", and "against" not preceded by "directly" encompass "directly against" as well as construction where intervening material(s), region(s), or structure(s) result(s) in no physical touching contact of the stated

materials, regions, or structures relative one another.

(41) Herein, regions-materials-components are "electrically coupled" relative one another if in normal operation electric current is capable of continuously flowing from one to the other and does so predominately by movement of subatomic positive and/or negative charges when such are sufficiently generated. Another electronic component may be between and electrically coupled to the regions-materials-components. In contrast, when regions-materials-components are referred to as being "directly electrically coupled", no intervening electronic component (e.g., no diode, transistor, resistor, transducer, switch, fuse, etc.) is between the directly electrically coupled regions-materials-components.

(42) Any use of "row" and "column" in this document is for convenience in distinguishing one series or orientation of features from another series or orientation of features and along which components have been or may be formed. "Row" and "column" are used synonymously with respect to any series of regions, components, and/or features independent of function. Regardless, the rows may be straight and/or curved and/or parallel and/or not parallel relative one another, as may be the columns. Further, the rows and columns may intersect relative one another at 90° or at one or more other angles (i.e., other than the straight angle).

(43) The composition of any of the conductive/conductor/conducting materials herein may be conductive metal material and/or conductively-doped semiconductive/semiconductor/semiconducting material. "Metal material" is any one or combination of an elemental metal, any mixture or alloy of two or more elemental metals, and any one or more metallic compound(s).

(44) Herein, any use of "selective" as to etch, etching, removing, removal, depositing, forming, and/or formation is such an act of one stated material relative to another stated material(s) so acted upon at a rate of at least 2:1 by volume. Further, any use of selectively depositing, selectively growing, or selectively forming is depositing, growing, or forming one material relative to another stated material or materials at a rate of at least 2:1 by volume for at least the first 75 Angstroms of depositing, growing, or forming.

(45) Unless otherwise indicated, use of "or" herein encompasses either and both.

Conclusion

(46) In some embodiments, a method used in forming memory circuitry comprises forming a stack comprising vertically-alternating first tiers and second tiers. The stack extends from a memory-array region into a stair-step region. The stair-step region comprises a cavity comprising a flight of stairs. The first tiers are conductive and the second tiers are insulative at least in a finished-circuitry construction. A lining is formed in and that less-than-fills the cavity atop treads of the stairs. Individual of the treads comprise conducting material of one of the first tiers in the finished-circuitry construction. The lining that is atop the treads is replaced with at least one of metal material, polysilicon, or SiGe and insulative material is provided in remaining volume of the cavity directly above the at least one of the metal material, the polysilicon, or the SiGe. Conductive vias are formed through the insulative material and the at least one of the metal material, the polysilicon, or the SiGe. Individual of the conductive vias are directly above and directly against the conducting material of one of the individual treads.

(47) In some embodiments, a method used in forming memory circuitry comprises forming a stack comprising vertically-alternating first tiers and second tiers. The stack extends from a memory-array region into a stair-step region. The stair-step region comprises a cavity comprising a flight of stairs. The first tiers are conductive and the second tiers are insulative at least in a finished-circuitry construction. A lining is formed in and that less-than-fills the cavity atop treads of the stairs. The treads individually comprise conducting material of one of the first tiers in the finished-circuitry construction. Insulative material is formed in remaining volume of the cavity directly above the lining. Through-array-via (TAV) openings are formed through the insulative material and the treads. Multiple of the TAV openings per tread extend through individual of the treads. Through the

TAV openings, the lining is isotropically etched laterally-outward relative to the TAV openings to horizontally interconnect the multiple TAV openings into a single opening per tread where the etched lining was. Through the TAV openings, at least one of metal material, polysilicon, or SiGe is formed in the single opening where the etched lining was and across horizontal locations where conductive vias to the conducting material of the individual treads will be formed. Conductive via openings are first etched through the insulative material in the horizontal locations to and using the at least one of the metal material, the polysilicon, or the SiGe as an etch-stop during the first etching such that the conductive via openings stop atop or within the at least one of the metal material, the polysilicon, or the SiGe. After the first etching, second etching is conducted through the at least one of the metal material, the polysilicon, or the SiGe to extend the conductive via openings deeper and the conducting material of the individual treads is exposed. Conductive vias are formed in the extended conductive via openings. Individual of the conductive vias are directly above and directly against the conducting material of one of the individual treads.

(48) In some embodiments, a method used in forming memory circuitry comprises forming a stack comprising vertically-alternating first tiers and second tiers. The first tiers comprise silicon nitride. The stack extends from a memory-array region into a stair-step region. The stair-step region comprises a cavity comprising a flight of stairs. The first tiers are conductive and the second tiers are insulative at least in a finished-circuitry construction. A lining is formed in and that less-than-fills the cavity atop treads of the stairs. The lining comprises silicon nitride. The treads individually comprise conducting material of one of the first tiers in the finished-circuitry construction. Insulative material is formed in remaining volume of the cavity directly above the lining. Through-array-via (TAV) openings are formed through the insulative material and the treads. Multiple of the TAV openings per tread extend through individual of the treads. Through the TAV openings, the silicon nitride of the lining and the silicon nitride of the first tiers laterally-outward relative to the TAV openings are isotropically etched. The silicon nitride of the lining is so laterally etched to a greater degree than is the silicon nitride of the first tiers. The isotropic etching horizontally interconnects the multiple TAV openings into a single opening per tread where the etched silicon nitride of the lining was. Through the TAV openings, at least one of metal material, polysilicon, or SiGe is formed in the single opening where the etched silicon nitride of the lining was and across horizontal locations where conductive vias to the conducting material of the individual treads will be formed. Conductive via openings are first etched through the insulative material in the horizontal locations and using the at least one of the metal material, the polysilicon, or the SiGe as an etch-stop during the first etching such that the conductive via openings stop atop or within the at least one of the metal material, the polysilicon, or the SiGe. After the first etching, second etching is conducted through the at least one of the metal material, the polysilicon, or the SiGe to extend the conductive via openings deeper and exposes the conducting material of the individual treads. Conductive vias are formed in the extended conductive via openings. Individual of the conductive vias are directly above and directly against the conducting material of one of the individual treads.

(49) In some embodiments, memory circuitry comprising strings of memory cells comprises a stack comprising vertically-alternating insulative tiers and conductive tiers. Channel-material strings of memory cells extend through the insulative tiers and the conductive tiers in a memory-array region. The insulative tiers and the conductive tiers extend from the memory-array region into a stair-step region. The stair-step region comprises a cavity comprising a flight of stairs. Treads of individual of the stairs comprise conducting material of one of the conductive tiers. Individual of the treads comprise at least one of metal material, polysilicon, or SiGe. The conducting material of one of the conductive tiers is directly below the at least one of the metal material, the polysilicon, or the SiGe. An intervening insulating material is vertically-between the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one conductive tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe. Insulative material is in the cavity directly above the treads. Conductive vias extend through the insulative

material. Individual of the conductive vias extend through the at least one of the metal material, the polysilicon, or the SiGe and the intervening insulating material to be directly against the conducting material of one of the individual treads.
(50) In compliance with the statute, the subject matter disclosed herein has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the claims are not limited to the specific features shown and described, since the means herein disclosed comprise example embodiments. The claims are thus to be afforded full scope as literally worded, and to be appropriately interpreted in accordance with the doctrine of equivalents.

Claims

1. A method used in forming memory circuitry, comprising: forming a stack comprising vertically-alternating first tiers and second tiers, the stack extending from a memory-array region into a stair-step region, the stair-step region comprising a cavity comprising a flight of stairs, the first tiers being conductive and the second tiers being insulative at least in a finished-circuitry construction; forming a lining in and that less-than-fills the cavity atop treads of the stairs, individual of the treads comprising conducting material of one of the first tiers in the finished-circuitry construction; replacing the lining that is atop the treads with at least one of metal material, polysilicon, or SiGe and providing insulative material in remaining volume of the cavity directly above the at least one of the metal material, the polysilicon, or the SiGe; and forming conductive vias through the insulative material and the at least one of the metal material, the polysilicon, or the SiGe; individual of the conductive vias being directly above and directly against the conducting material of one of the individual treads.
2. The method of claim 1 wherein the providing of the insulative material in the remaining volume of the cavity occurs before the replacing of the lining with the at least one of the metal material, the polysilicon, or the SiGe.
3. The method of claim 1 wherein the at least one comprises metal material.
4. The method of claim 1 wherein the at least one comprises polysilicon.
5. The method of claim 1 wherein the at least one comprises SiGe.
6. The method of claim 1 wherein the forming of the conductive vias comprises: first etching conductive via openings through the insulative material for the conductive vias and using the at least one of the metal material, the polysilicon, or the SiGe as an etch-stop during the first etching such that the conductive via openings stop atop or within the at least one of the metal material, the polysilicon, or the SiGe; after the first etching, second etching through the at least one of the metal material, the polysilicon, or the SiGe to extend the conductive via openings deeper and exposing the conducting material of the individual treads; and forming the individual conductive vias in individual of the extended conductive via openings.
7. The method of claim 1 wherein the individual treads in the finished-circuitry construction comprise: the at least one of the metal material, the polysilicon, or the SiGe; the conducting material of one of the first tier directly below the at least one of the metal material, the polysilicon, or the SiGe; and an intervening insulating material vertically-between the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one first tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe.
8. The method of claim 7 wherein the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one first tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe are of different compositions relative one another.
9. The method of claim 8 wherein the at least one of the metal material, the polysilicon, or the SiGe is conductive.
10. The method of claim 9 wherein the at least one of the metal material, the polysilicon, or the

SiGe and the conducting material of the one first tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe are of different compositions relative one another.

11. The method of claim 9 wherein the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one first tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe are of the same composition relative one another.

12. The method of claim 7 wherein the at least one of the metal material, the polysilicon, or the SiGe is insulative.

13. The method of claim 12 wherein the at least one of the metal material, the polysilicon, or the SiGe comprises Al.sub.2O.sub.3.

14. The method of claim 7 comprising forming a through-array-via (TAV) construction extending through the insulative material, through the at least one of the metal material, the polysilicon, or the SiGe, through the intervening insulating material, and through the conducting material of the one first tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe of the individual treads.

15. The method of claim 14 comprising multiple of the TAV constructions per individual tread.

16. The method of claim 15 wherein the at least one of the metal material, the polysilicon, or the SiGe is conductive and is horizontally-continuous among the multiple TAV constructions in the respective individual tread.

17. A method used in forming memory circuitry, comprising: forming a stack comprising vertically-alternating first tiers and second tiers, the stack extending from a memory-array region into a stair-step region, the stair-step region comprising a cavity comprising a flight of stairs, the first tiers being conductive and the second tiers being insulative at least in a finished-circuitry construction; forming a lining in and that less-than-fills the cavity atop treads of the stairs, the treads individually comprising conducting material of one of the first tiers in the finished-circuitry construction; forming insulative material in remaining volume of the cavity directly above the lining; forming through-array-via (TAV) openings through the insulative material and the treads, multiple of the TAV openings per tread extending through individual of the treads; through the TAV openings, isotropically etching the lining laterally-outward relative to the TAV openings to horizontally interconnect the multiple TAV openings into a single opening per tread where the etched lining was; through the TAV openings, forming at least one of metal material, polysilicon, or SiGe in the single opening where the etched lining was and across horizontal locations where conductive vias to the conducting material of the individual treads will be formed; first etching conductive via openings through the insulative material in the horizontal locations to and using the at least one of the metal material, the polysilicon, or the SiGe as an etch-stop during the first etching such that the conductive via openings stop atop or within the at least one of the metal material, the polysilicon, or the SiGe; after the first etching, second etching through the at least one of the metal material, the polysilicon, or the SiGe to extend the conductive via openings deeper and exposing the conducting material of the individual treads; and forming conductive vias in the extended conductive via openings, individual of the conductive vias being directly above and directly against the conducting material of one of the individual treads.

18. A method used in forming memory circuitry, comprising: forming a stack comprising vertically-alternating first tiers and second tiers, the first tiers comprising silicon nitride, the stack extending from a memory-array region into a stair-step region, the stair-step region comprising a cavity comprising a flight of stairs, the first tiers being conductive and the second tiers being insulative at least in a finished-circuitry construction; forming a lining in and that less-than-fills the cavity atop treads of the stairs, the lining comprising silicon nitride, the treads individually comprising conducting material of one of the first tiers in the finished-circuitry construction; forming insulative material in remaining volume of the cavity directly above the lining; forming through-array-via (TAV) openings through the insulative material and the treads, multiple of the TAV openings per tread extending through individual of the treads; through the TAV openings,

isotropically etching the silicon nitride of the lining and the silicon nitride of the first tiers laterally-outward relative to the TAV openings; the silicon nitride of the lining being so laterally etched to a greater degree than is the silicon nitride of the first tiers, the isotropically etching horizontally interconnecting the multiple TAV openings into a single opening per tread where the etched silicon nitride of the lining was; through the TAV openings, forming at least one of metal material, polysilicon, or SiGe in the single opening where the etched silicon nitride of the lining was and across horizontal locations where conductive vias to the conducting material of the individual treads will be formed; first etching conductive via openings through the insulative material in the horizontal locations and using the at least one of the metal material, the polysilicon, or the SiGe as an etch-stop during the first etching such that the conductive via openings stop atop or within the at least one of the metal material, the polysilicon, or the SiGe; after the first etching, second etching through the at least one of the metal material, the polysilicon, or the SiGe to extend the conductive via openings deeper and exposing the conducting material of the individual treads; and forming conductive vias in the extended conductive via openings, individual of the conductive vias being directly above and directly against the conducting material of one of the individual treads.
19. Memory circuitry comprising strings of memory cells, comprising: a stack comprising vertically-alternating insulative tiers and conductive tiers, channel-material strings of memory cells extending through the insulative tiers and the conductive tiers in a memory-array region; the insulative tiers and the conductive tiers extending from the memory-array region into a stair-step region, the stair-step region comprising a cavity comprising a flight of stairs, treads of individual of the stairs comprising conducting material of one of the conductive tiers, individual of the treads comprising: at least one of metal material, polysilicon, or SiGe; the conducting material of one of the conductive tiers directly below the at least one of the metal material, the polysilicon, or the SiGe; and an intervening insulating material vertically-between the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one conductive tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe; insulative material in the cavity directly above the treads; and conductive vias extending through the insulative material, individual of the conductive vias extending through the at least one of the metal material, the polysilicon, or the SiGe and the intervening insulating material to be directly against the conducting material of one of the individual treads.
20. The memory circuitry of claim 19 wherein the at least one comprises metal material.
21. The memory circuitry of claim 19 wherein the at least one comprises polysilicon.
22. The memory circuitry of claim 19 wherein the at least one comprises SiGe.
23. The memory circuitry of claim 19 wherein the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one conductive tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe are of different compositions relative one another.
24. The memory circuitry of claim 19 wherein the at least one of the metal material, the polysilicon, or the SiGe is conductive.
25. The memory circuitry of claim 24 wherein the at least one of the metal material, the polysilicon, or the SiGe comprises an elemental metal or an alloy of two or more elemental metals.
26. The memory circuitry of claim 24 wherein the at least one of the metal material, the polysilicon, or the SiGe comprises a metallic compound.
27. The memory circuitry of claim 24 wherein the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one conductive tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe are of different compositions relative one another.
28. The memory circuitry of claim 24 wherein the at least one of the metal material, the polysilicon, or the SiGe and the conducting material of the one conductive tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe are of the same

composition relative one another.
29. The memory circuitry of claim 24 wherein the at least one of the metal material, the polysilicon, or the SiGe is not directly against the individual conductive via extending there-through.
30. The memory circuitry of claim 19 comprising a through-array-via (TAV) construction extending through the insulative material, through the at least one of the metal material, the polysilicon, or the SiGe, through the intervening insulating material, and through the conducting material of the one conductive tier that is directly below the at least one of the metal material, the polysilicon, or the SiGe of the individual treads.
31. The memory circuitry of claim 30 comprising multiple of the TAV constructions in individual of the individual treads.
32. The memory circuitry of claim 31 wherein the at least one of the metal material, the polysilicon, or the SiGe is conductive and is horizontally-continuous among the multiple TAV constructions in the individual treads.
33. The memory circuitry of claim 19 wherein the at least one of the metal material, the polysilicon, or the SiGe is insulative.
34. The memory circuitry of claim 33 wherein the at least one of the metal material, the polysilicon, or the SiGe comprises Al.sub.2O.sub.3.